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Submitted via email to AirCleaners2021STD0035@ee.doe.gov and AirCleaners2021TP0036@ee.doe.gov

April 10, 2022

Dr. Stephanie Johnson
U.S. Department of Energy
Office of Energy Efficiency and Renewable Energy
Building Technologies Office
EE-5B, 1000 Independence Avenue, SW
Washington, DC 20585-0121

RE: Request for Comments Regarding DOE's Request for Information for Energy Conservation Program: Test Procedure and Energy Conservation Standards for Consumer Products; Consumer Air Cleaners [Docket # EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036]

Dear Dr. Johnson:

Daikin U.S. Corporation ("Daikin") is respectfully submitting the following comments in response to the U.S. Department of Energy's ("DOE") Request for Information ("RFI") published in the Federal Register, Vol. 87, No. 16 on January 25, 2022, seeking input regarding the Test Procedure and Energy Conservation Standards for "air cleaners."

Daikin U.S. is a subsidiary of Daikin Industries, Ltd., the world's largest heating, ventilation and air conditioning ("HVAC") equipment manufacturer. Daikin U.S. is headquartered in Washington D.C. and represents various Daikin Group companies--including Daikin Comfort Technologies North America, Inc. formerly Goodman Manufacturing Company, L.P. ("DCT"), American Air Filter Company, Inc. ("AAF") and Daikin Applied Americas Inc. ("DAA")--that manufacture residential, light commercial, commercial heating and cooling equipment, and refrigeration equipment sold and installed by contractors in every American state, as well as in Canada.

1. Definition, scope and product classification

Daikin understands DOE's intent to regulate air cleaners due to their growing popularity. However, this Request for Information for test procedures and energy conservation standard is premature. Instead DOE should further deliberate on the industry's comments made during the rulemaking procedure for Notice of Proposed Determination (NOPD). Daikin believes that more clarity is needed related to scope and definition of "air cleaner" before DOE can consider applicable test procedures and relevant energy conservation standards. DOE will be better informed if it first attempts to classify the variety of air cleaners in the market into manageable categories and then prioritize the air cleaner categories that present the largest scope for energy savings and technological advancement.

DOE's intended definition of air cleaners may sufficiently address the primary function of the equipment; however, it is not appropriate when applied with the intent of developing test procedures or energy conservation standards. Similar to HVAC systems that can be defined by their function of providing heating, cooling or ventilation, air cleaners need to be classified in different categories based on the method of providing the said function, capacity of applications, types of occupancies served in order to develop meaningful test procedures or energy conservation standards. DOE's current approach creates a broad equipment class for air cleaners within which it includes a variety of air cleaning equipment that cannot be represented by a single performance rating or metric, nor tested in the same way. A one size fits all approach is not appropriate here.

Consequently, Daikin cannot answer the myriad of DOE's questions in any meaningful way before understanding what DOE wants to focus on or prioritize. DOE's latest consideration to expand the originally proposed definition to include consumer air cleaners, "such as those that are mounted on walls and ceilings, or that are designed for whole-home air cleaning in conjunction with central heating or air conditioning systems" further minimizes the possibility of a meaningful response to its request for information. AHAM and EPA have developed programs specifically for air cleaners that can be moved from room-to-room and the test procedure is only applicable to those type of products. The limitations of the test procedure, whether intentionally or inadvertently, also creates a clear distinction between products that are truly consumer products available to be bought at retail or online stores. It also excludes those air cleaners that are used commercially, for example, in hospitals or office buildings, simply based on their capacity or configuration. Daikin urges DOE to review its recommended approach as explained in our November 15, 2021 comments to the NOPD (see Appendix A). DOE's intent may be to exclude commercial products in the scope of the rulemaking, but unfortunately, the definition does not reflect that. DOE must at least create categories of air cleaners based on their capacity, configuration and air cleaning technology before determining test procedures. We recommend that DOE adopt a maximum airflow of 400 CFM to differentiate between residential and commercial equipment. Daikin can provide responses to DOE's RFI questions based on DOE's clarification to scope and product classification.

2. **Test procedures and performance metrics**

DOE is correct in identifying the AHAM test procedure for rating some specific performance metrics of an air cleaner. However, the performance metrics and related test procedure applies to portable small room air cleaners only. It can only measure the amount of clean air delivered (cfm) without measuring the efficacy of air cleaning technology. As a result, a HEPA filter can be penalized over a MERV 8 filter, despite having superior air cleaning capabilities. Consequently, a presumably efficient product with MERV 8 filter will be ineffective for residences in a polluted area and other similar applications. This further illustrates the need to create meaningful product categories.

Daikin encourages DOE to explore test procedures from other global markets in Asia and Europe.

DOE's assessment of performance metrics could be expanded to metrics other than a clean air delivery rate (CADR). CADR primarily measures the capacity of the unit, however, there are other air cleaning efficacy metrics that should be considered based on product categories. Metrics like CADR and MERV are similar to the capacity of delivering clean air and air cleaning efficacy respectively; they are not an energy efficiency metric. EPCA requires that a test procedure must be reasonably designed to produce test results which reflect energy efficiency, energy use or estimated annual operating cost of a given type of covered product during a representative average use cycle and not be unduly burdensome to conduct.

3. **Operation and usage profile**

One of the key tasks in determining need for regulation, and subsequently the energy consumption of air cleaners via test procedures, is operation and usage profile. DOE's assumption that an air cleaner runs at 100% capacity for 16 hours of the day is flawed. Based on a quick survey[1] of portable air cleaners available in the market today, it is evident that most, if not all, air cleaners have multiple modes of operation. Most

[1] Honeywell InSight HEPA Air Purifier - Black, HPA5100B, https://www.honeywellstore.com/store/products/honeywell-insight-series-hepa-air-purifier-hpa5100b.htm, (last visited April 10, 2022)
Kenmore 83395 Medium Room HEPA Filter Air Purifier, https://www.kenmore.com/products/kenmore-83395-medium-room-hepa-filter-air-purifier/ (last visited April 10, 2022)
GermGuardian AC4825 22" Air Purifier with HEPA & UVC, https://www.guardiantechnologies.com/collections/air-purifiers-with-uv-hepa-filters/products/ac4825-4-in-1-air-cleaning-system-with-true-hepa-uv-c-sanitizer-and-odor-reduction-22-inch-tower-by-germguardian (last visited April 10, 2022)
Levoit Core 300S Smart True HEPA Air Purifier, https://levoit.com/products/core-300s-smart-true-hepa-air-purifier (last visited April 10, 2022)
Blue Air Protect 7410i, https://www.blueair.com/us/protect/protect-7410i/2207.html (last visited April 10, 2022)

air cleaners are recommended to run on auto-mode, which runs the unit at 100% capacity only when indoor air quality (IAQ) drops. Higher capacity operations are only required during events of poor IAQ, which are arguably only occur a few times in residential applications. While this strategy may also apply to ceiling or wall mounted air cleaners, the operation profile can be drastically different for ducted or in-duct air cleaners.

4. **DOE's rulemaking process**

Since this is a new product rulemaking, DOE must strictly follow its processes in its appropriate and intended order. DOE must finish its determination and then commence a test procedure rule, and finalize it, before establishing an energy conservation standard. This is important because first and foremost, DOE must provide clarity to the product categories and what it intends to regulate. Once that is determined, the industry can engage in a meaningful discussion around appropriate test procedures, as there may be different test procedures for the different types/classes of air cleaners. And because of the fast evolution of air cleaners, DOE must afford the industry sufficient time to comply with test procedures before determining minimum energy efficiency and other relevant performance standards. We believe that there may be laboratory test chamber shortages during and after DOE determines test procedures.

5. **Conclusion**

Daikin appreciates the opportunity to provide these comments. If you have any questions regarding this submission, please do not hesitate to contact myself or Rusty Tharp, Vice President, Regulatory Affairs and Environmental Research at Daikin Comfort Technologies (rusty.tharp@daikincomfort.com).

Respectfully submitted,

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Appendix A
Daikin's Letter of Comments to DOE's NOPD on Air Cleaners

Submitted via email to AirCleaners2021DET0022@ee.doe.gov

November 15, 2021

Dr. Stephanie Johnson
U.S. Department of Energy
Office of Energy Efficiency and Renewable Energy
Building Technologies Office
EE-5B, 1000 Independence Avenue, SW
Washington, DC 20585-0121

RE: Request for Comments Regarding DOE's Notification of Proposed Determination of Air Cleaners as a Covered Consumer Product [Docket # EERE-2021-BT-DET-0022]

Dear Dr. Johnson:

Daikin U.S. Corporation ("Daikin") is respectfully submitting the following comments in response to the U.S. Department of Energy's ("DOE") Request for Comments ("RFC") published in the Federal Register, 86 Fed. Reg. No. 177 (September 16, 2021) and seeking input regarding the coverage of "air cleaners."

Daikin U.S. is a subsidiary of Daikin Industries, Ltd., the world's largest heating, ventilation and air conditioning ("HVAC") equipment manufacturer. Daikin U.S. is headquartered in Washington D.C. and represents various Daikin Group companies--including Goodman Manufacturing Company, L.P. ("Goodman"), American Air Filter Company, Inc. ("AAF") and Daikin Applied Americas Inc. ("DAA")--that manufacture residential, light commercial, commercial heating and cooling equipment, and refrigeration equipment sold and installed by contractors in every American state, as well as in Canada.

6. Scope and Definition

Daikin U.S. understands DOE's intent to regulate air cleaners due to their growing popularity. The request for information in the current Notice of Proposed Determination (NOPD) provides a starting point for related discussions, but Daikin U.S. believes that more clarity is needed related to scope and definition of "air cleaner" before making the determination that air cleaners should be regulated.

> From the EPCA's consumer product definition[2], we believe that covered products would be only those air cleaners that are marketed and sold to consumers for use in residential applications. The proposed definition is overly broad and could encapsulate unintended products. As such, in considering the potential scope of coverage, DOE does not consider whether an individual product

[2] "(B) which, to any significant extent, is distributed in commerce for personal use or consumption by individuals; without regard to whether such article of such type is in fact distributed in commerce for personal use or consumption by an individual." (42 U.S.C. 6291(a)(1)).

is distributed in commerce for residential or commercial use, but whether it is of a type of product distributed in commerce for residential use. We believe the definition should exclude any products marketed and sold solely for commercial applications. Some air cleaners are clearly marketed and sold solely for commercial applications, such as AAF's *AstroPure*[3] and DAA's *DAP* products.[4] These products are typically not acquired through typical retail stores, but rather via a contractor, dealer or distributor.

- DOE proposes a definition and product description in the NOPD based on AHAM and ENERGY STAR qualified products. However, DOE's proposed definition can potentially cover a broader range of products that cannot be tested using AHAM's standard test procedure (AHAM AC-1 2020, Method for Measuring Performance of Portable Household Electric Room Air Cleaners) or ENERGY STAR specification. The scope of products in AHAM's test procedure are portable air cleaners that can be placed in the center of the test room, verify Clean Air Delivery Rate (CADR) ratings up to 450 CFM (pollen) and -600 CFM (dust and cigarette smoke)[5]. CADR, however, is a performance metric that is only discovered after testing. Consequently, a classification of equipment must rely on the nominal airflow of the unit.
- Additionally, a consumer air cleaner placed in a residential space, with existing HVAC equipment will seldom require a high CFM. Most residential room air cleaners have relatively low airflow rates. E.g. A 400 CFM unit can achieve 2 ACH per ASHRAE recommendation for space as large as 2400 ft^2 space, when assuming a 10 ft ceiling height. This is approximately 48x48 ft room size, which generally does not exist in residential spaces. Consequently, an air cleaner capacity of 400 CFM and above can be considered excessive and unnecessary.
- Lastly, some products whose primary function may be humidification or dehumidification may, as a secondary function, also filter air. Although these are separate functions, the mechanism to generate airflow is the same and an energy consumption measurement cannot be isolated to air cleaning only.
- Based on the above factors and DOE's intent to cover consumer products only, we recommend DOE consider creating a clearer definition for "consumer product" air cleaners. Daikin recommends revisions to DOE's proposed definition in the NOPD, shown below with track changes:

    Air cleaner is defined as a consumer product that:

    (1) Is a self-contained, mechanically encased assembly; and
    (2) Is directly powered by 120 V, single-phase electric current supplied by a plug; and
    (3) Has a maximum airflow rate of 400 CFM;
    (4) Removes, destroys, or deactivates particulates and microorganisms from the air;
    (5) Is marketed for residential use for the primary functions of removing, destroying, or deactivating particulates and microorganisms from the air, and
    (6) Excludes:

    (i) Products that destroy or deactivate particulates and microorganisms solely by means of ultraviolet light or electrostatic air filters without a fan for air circulation; ~~and~~ or
    (ii) Products without a fan; or
    (iii) products that provide incidental air cleaning as a result of performing their other main functions, such as humidification, dehumidification, refrigeration; or

[3] https://www.aafintl.com/en/commercial/browse-products/commercial/portable-air-purification-systems/astropure. "Typical Applications: - Healthcare, - Schools and Universities, -Government Infrastructure/Homeland Security."
[4] https://www.daikinapplied.com/products/airpurifier – "within indoor spaces including classrooms, dental offices, clinics, office spaces, churches, restaurants, and hospitals."
[5] Maximum CADR in AHAM-AC-1-2015 was 450 CFM for cigarette smoke, dust and pollen.

(iv) products powered by external transformer(s), or
(v) ancillary products used in conjunction with, or inside ducts of, HVAC equipment;
(vi) central air conditioners, room air conditioners, portable air conditioners, dehumidifiers, and furnaces as defined in 10 CFR 430.2.

7. Technology

- *DOE seeks comment on availability or lack of availability of technologies for improving energy efficiency of air cleaners.* Daikin believes there may be challenges in further improving energy efficiency of air cleaners, as the majority of potentially covered products, to our understanding of consumer product air cleaners, are free-air discharge for which air-moving technology does not have significant potential gains in energy efficiency to our knowledge. We do not believe significant gains are feasible with changes in motors and impellers, or modifications to reduce energy consumption in stand-by and off-mode. Reduction in energy consumption by reducing pressure drop due to filters would only harm the air cleaning efficiency of the products.

8. Average Annual Per-Household Energy Use

- *Data and information regarding annual energy use estimates for air cleaners, particularly for products not covered by the ENERGY STAR program, such as non-portable products (wall-mounted, ceiling-mounted, and whole-home units.* Daikin understands DOE's approach to analyze Annual Energy Use based on the information available via EPA's ENERGY STAR listing, however, it is not clear how Annual Energy Use is calculated. Specifically, it should consider number of operating hours, which is not clear in the AHAM test procedure or from EPA's data. Daikin requests that DOE disclose its methodology and results of Annual Energy Use assessment. A similar methodology could then be applied to other products not covered by ENERGY STAR program.
- The actual hours of operation will obviously have a significant impact on the annual energy consumption of a product. We do not have specific information as to typical operational hours of consumer/residential air cleaners.

9. Labeling

- DOE notes in the NOPD that one criteria for determining a product to be a covered product is the "application of a labeling rule under 42 U.S.C. 6294 is unlikely to be sufficient to induce manufacturers to produce, and consumers and other persons to purchase, covered products of such type (or class) that achieve the maximum energy efficiency that is technologically feasible and economically justified." (42 U.S.C. 6295(l)(1)).
- Our opinion is that labeling could be a significant factor in driving manufacturer and consumer behavior. We believe it is generally accepted that food nutritional and content labeling has significantly impacted consumer behavior on food purchases. One study from earlier days of consumer nutritional labeling found that "almost 48% of consumers reported that they changed their purchasing behavior due to nutritional labels."[6] While there is a wide variety of research as to the effectiveness of energy efficiency labeling, one report on the psychology of labeling[7] indicates that energy efficiency labeling can be an effective tool to influence consumer behavior,

[6] https://www.researchgate.net/publication/325827891_Food_Label_and_its_influence_on_Consumer_Buying_Behavior_A_Review_of_Research_Studies.
[7] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8558512/.

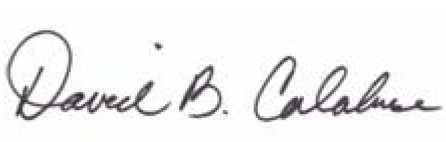
at least in part based on label trust. The Federal Trade Commission Energy Guide is label that is generally trusted by the American people.

- We suggest that with the relative energy consumption of these products, and the above information, that air cleaners as a consumer product would be a good product for a labeling rule to drive manufacturer and consumer behavior to higher efficiency products.

10. <u>Conclusion</u>

Daikin appreciates the opportunity to provide these comments. If you have any questions regarding this submission, please do not hesitate to contact myself or Rusty Tharp, Senior Director of Regulatory Affairs at Goodman (rusty.tharp@goodmanmfg.com).

Respectfully submitted,



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